

Indicator Transforms

“Now I see,” said Tom blindly.

An indicator transform is a means to shape the indicator to aid interpretation and use of the indicator itself. Other than shaping the display, these transforms have no impact on the indicator. For example, they induce no lag. While relatively trivial to implement, these transforms have a huge impact on the interpretation and use of the indicators. There are primarily three indicators of interest to traders: the Fisher transform, the inverse Fisher transform, and the cube transform. Each has its preferred application, which we describe in this chapter.

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■ Fisher Transform

A common fallacy in the use of technical indicators is that their probability distribution follows a normal, or Gaussian, probability distribution. That just is not so, as you can prove for yourself by analyzing a reasonable amount of data. If the data do not have a normal probability distribution, then the calculation of a standard deviation is based on a false premise. The purpose of the Fisher transform is to take any indicator having a nominally zero mean and bounded between the limits of -1 to $+1$ and convert the amplitude so that the transformed indicator has an approximate normal probability distribution.

Plotting an indicator scaled to a normal probability distribution has a huge advantage on indicator interpretation. If the transformed indicator has a value of -1 , it has a value of negative one standard deviation and therefore

there is a 32 percent chance prices will go lower. If the transformed indicator has a value of -2 , it has a value of negative two standard deviations, and therefore there is only an 8 percent chance prices will go lower. This is a high-probability buying opportunity. At a level of -3 , the negative 3 standard deviations means there is only a 2 percent chance of the prices going lower. The transformed values are symmetrical, so positive deviations are high-probability indications to exit a long position or to sell short.

The equation for a Fisher transform is

$$\text{Output} = 0.5 * \ln \left[\frac{1 + \text{Input}}{1 - \text{Input}} \right]$$

Since an input value of unity can cause the ratio to go to infinity, the input must be limited to 0.999, which corresponds to an output on the order of three standard deviations. Similarly, the argument of the natural logarithm cannot be allowed to go to zero, and so the input must also be limited at -0.999 .

The transfer response of the Fisher transform is shown in Figure 15.1. The straight line shows the range of inputs, and the curved line shows the corresponding range of outputs. When the inputs are less than an absolute value of 0.5, the outputs are almost the same as the inputs. However, there is a greater and greater magnification of the outputs as the inputs approach the absolute value of 1. It is this nonlinear magnification that produces the “long tails” in the output probability distribution, thus creating the resulting normal probability distribution.

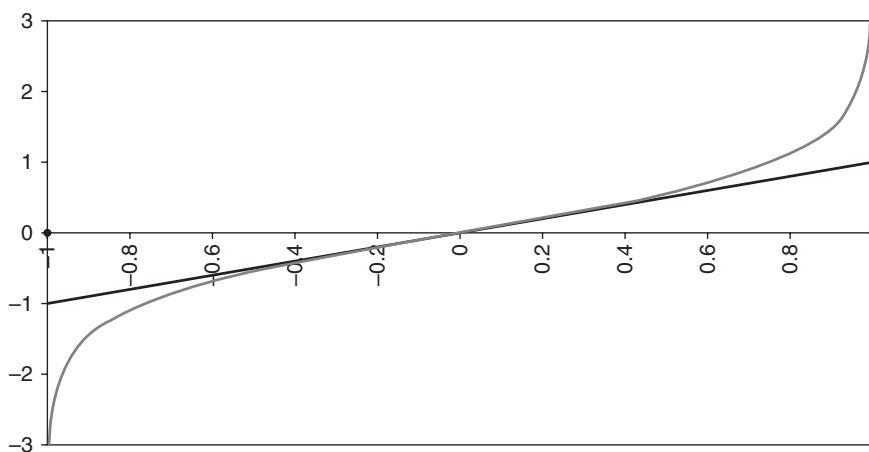


FIGURE 15.1 Transfer Response of the Fisher Transform Shows Greater Magnification of the Outputs as the Inputs Approach an Absolute Value of 1

To implement the Fisher transform on the adaptive RSI, replace the plot statements of Code Listing 11-1 with the following code fragment:

Code Listing 15-1. EasyLanguage Code Fragment to Add the Fisher Transform to the Adaptive RSI Indicator

```
Vars:
    TranslatedRSI(0),
    AmplifiedRSI(0),
    Fish(0);

TranslatedRSI = 2*(MyRSI - .5);
AmplifiedRSI = 1.5*TranslatedRSI;
If AmplifiedRSI > .999 Then AmplifiedRSI = .999;
If AmplifiedRSI < -.999 Then AmplifiedRSI = -.999;
Fish = .5*Log((1 + AmplifiedRSI) / (1 - AmplifiedRSI));
Plot1(Fish);
Plot2(2);
Plot6(-2);
```

The variable *MyRSI* ranges between zero and one, and therefore to accommodate the conditions for the Fisher transform, this variable must be translated and dilated to range between -1 and $+1$. If the *MyRSI* variable does not range fully between zero and one, you can shorten the RSI look-back period to be less than half the measured dominant cycle, or you can simply multiply it by a magnification factor as I have done in the code fragment. The amplifying factor was selected to make the indicator rarely exceed the two sigma points in the output. The amplified RSI is then limited to be within the range of -0.999 to $+0.999$ to avoid a computer crash, and then is used to compute the Fisher transform. The plus and minus two standard deviation levels are included in the indicator display. An example of the adaptive RSI using the Fisher transform is shown in Figure 15.2. The trend up is heralded by the Fisherized indicator's falling below -2 . Then, the trend down is signaled by the Fisherized indicator rising above $+2$. These major turning points are relatively rare events and are perhaps best used in association with trend trading.

Since the Stochastic and commodity channel index (CCI) indicators measure relative position in a channel, they tend to be near the upper or lower limits during trends. That means that they do not have a zero mean in the short run and therefore are not good candidates on which to apply the

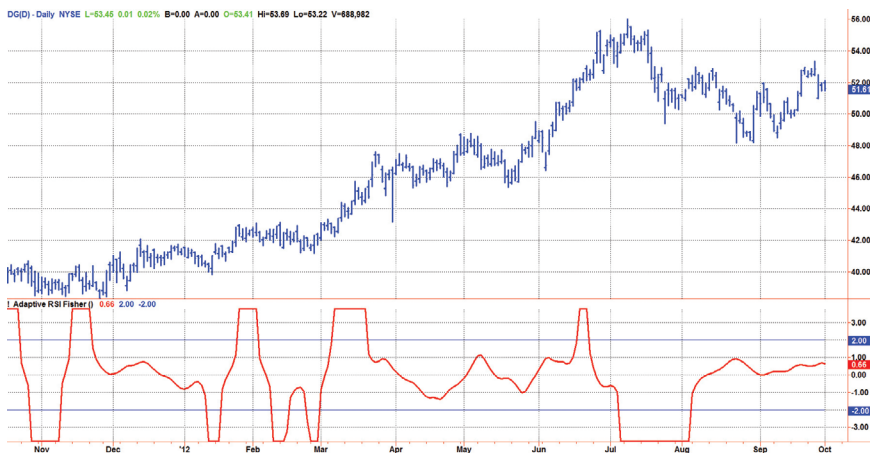


FIGURE 15.2 Adaptive RSI with Fisher Transform Clearly Identifies Major Turning Points

Fisher transform. A band-pass filter would be a much better candidate, but the major cyclic turning points are already easy to identify and don't require a more complex statistical approach.

■ Inverse Fisher Transform

Whereas the Fisher transform expands the base indicator near the extremes, the purpose of the inverse Fisher transform is to compress the indicator in these regions. By compressing values near the extremes many extraneous and irrelevant wiggles are removed from the indicator, making interpretation of the real meaning of the indicator be far simpler. In a sense, the inverse Fisher transform acts as a soft limiter.

The equation for the inverse Fisher transform is

$$\text{Output} = \frac{e^{(2*K*Input)} - 1}{e^{(2*K*Input)} + 1}$$

The input may have as large a value as desired. If the input values are very large, then the exponential terms in both the numerator and denominator are very large compared to one, and so the output is approximately 1. If the input values are very large, but negative, then the exponential terms in both the numerator and denominator are very small compared to one, with the result that the output is approximately -1. The K in the expression is an optional amplifying term because conventional indicators already fall between

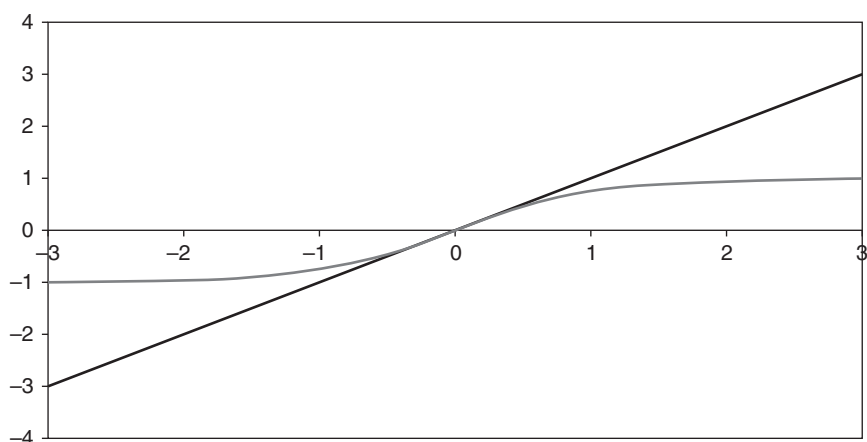


FIGURE 15.3 Transfer Response of the Inverse Fisher Transform Shows the Compression of Larger Values of the Input

the limits of -1 and $+1$, and so amplifying them is necessary to reap the benefits of the soft limiting of the inverse Fisher transform.

The transfer response of the inverse Fisher transform is shown in Figure 15.3. The input values are shown by the straight line, and the output values are shown by the curved line. This graphical representation shows how larger and larger absolute values of the input are compressed to essentially be unity.

To implement the inverse Fisher transform on an adaptive Stochastic indicator, replace the plot statements of Code Listing 11-2 with the code fragment given in Code Listing 15-2. I have chosen to amplify the adaptive Stochastic by a factor of 3 when taking the inverse Fisher transform.

Code Listing 15-2. EasyLanguage Code Fragment to Implement the Inverse Fisher Transform on the Adaptive Stochastic Indicator

```
Vars:
    IFish(0);

    Value1 = 2*(AdaptiveStochastic - .5);
    IFish = (ExpValue(2*3*Value1) - 1) / (ExpValue
        (2*3*Value1) + 1);
    Plot1(IFish);
    Plot4(.9*IFish[1]);
```

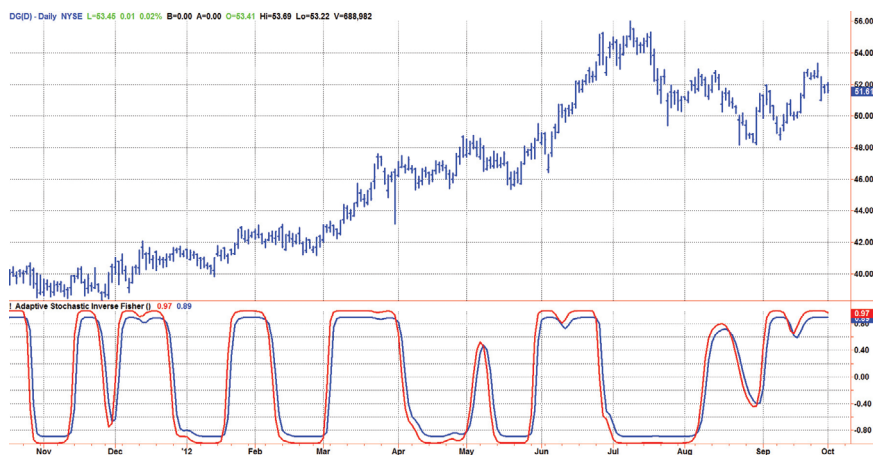


FIGURE 15.4 The Inverse Fisher Transform of the Adaptive Stochastic Indicator Gives Clear and Unambiguous Indications of the Proper Buy and Sell Points

I have also included a trigger line, which is just the inverse Fisher transform delayed by one bar and attenuated to 90 percent. The crossings of the inverse Fisher transform line and the trigger line provide clear and unequivocal indications for the buy and sell points, as demonstrated in the example of Figure 15.4.

The use of the inverse Fisher adaptive Stochastic indicator involves the crossings of the inverse Fisher line and the trigger line. When the inverse Fisher line crosses over the trigger line, then buy. When the inverse Fisher line crosses under the trigger line, then sell short, or if you choose, exit the long position.

The inverse Fisher transform should be one of the biggest weapons in your arsenal of trading indicators and strategies. Its use as a soft limiter is universal and can help remove many of the distracting and irrelevant squiggles in your indicators. The only constraint should be that the original input has a nearly a zero mean. You may need to amplify the input to attain the desired compression at the output. If you want a simple “digital” output, just amplify the input by a large amount.

■ Cube Transform

The purpose of the cube transform is to compress the signals near zero of an indicator that swings between the limits of -1 and $+1$. This is handy for an indicator such as the adaptive band-pass filter where you are concerned only with the larger swings, and the squiggles in the middle are merely distractions. The transform consists of merely cubing the indicator values.

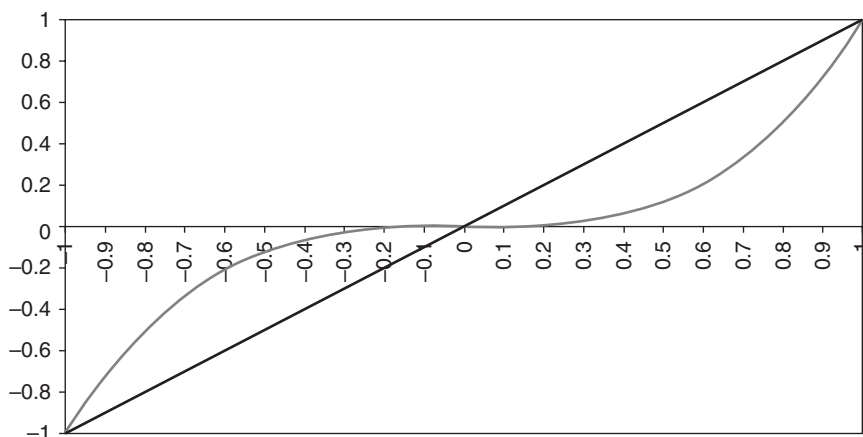


FIGURE 15.5 The Cube Transform Compresses Smaller Values of the Input

Doing this, the values near +1 and -1 are nearly unchanged, but the smaller absolute values are severely reduced in amplitude. Of course, this only works if the effects of spectral dilation have been removed by a roofing filter or some equivalent technique. The cubing of the display is only effective when the original indicator has a nominally zero mean. The transfer response of the cube transform is shown in Figure 15.5. In this figure, the input values are shown by the straight line and the output values are shown by the curved line.

The effect of the cube transform is shown in Figure 15.6, where the adaptive band-pass filter is shown in the first subgraph, and the adaptive

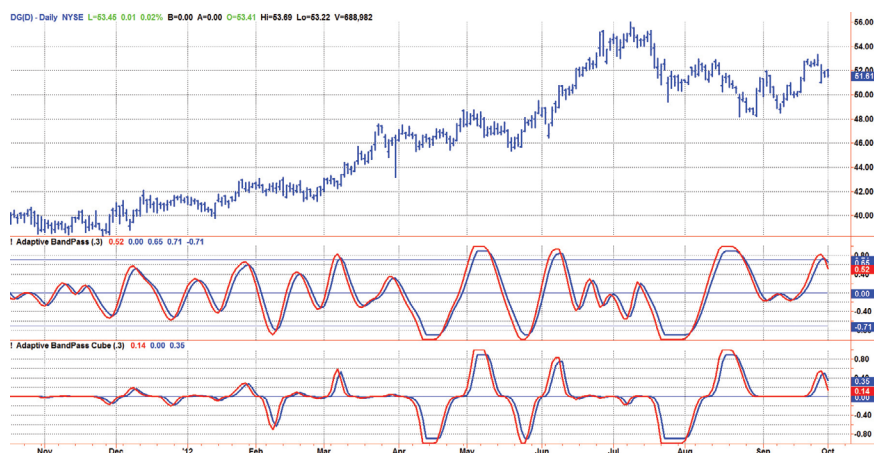


FIGURE 15.6 The Cube Transform Clarifies Swinging Signals by Compressing the Smaller Amplitude Swings

band-pass filter with the cube transform is shown in the second subgraph. The purpose of the cube transform is to clarify the major swings by compressing the distractions of the smaller swing values.

■ Key Points to Remember

1. The purpose of a transformer is shape the indicator to aid interpretation and use of the indicator itself.
2. Transformers do not induce lag.
3. A Fisher transform of an indicator swinging between -1 and $+1$ with a nominal zero mean plots that indicator in terms of standard deviations with a nearly normal probability distribution.
4. An inverse Fisher transform acts as a soft limiter to remove extraneous wiggles in indicators have a nominal zero mean.
5. A cube transform compresses the smaller values of an indicator swinging between -1 and $+1$.